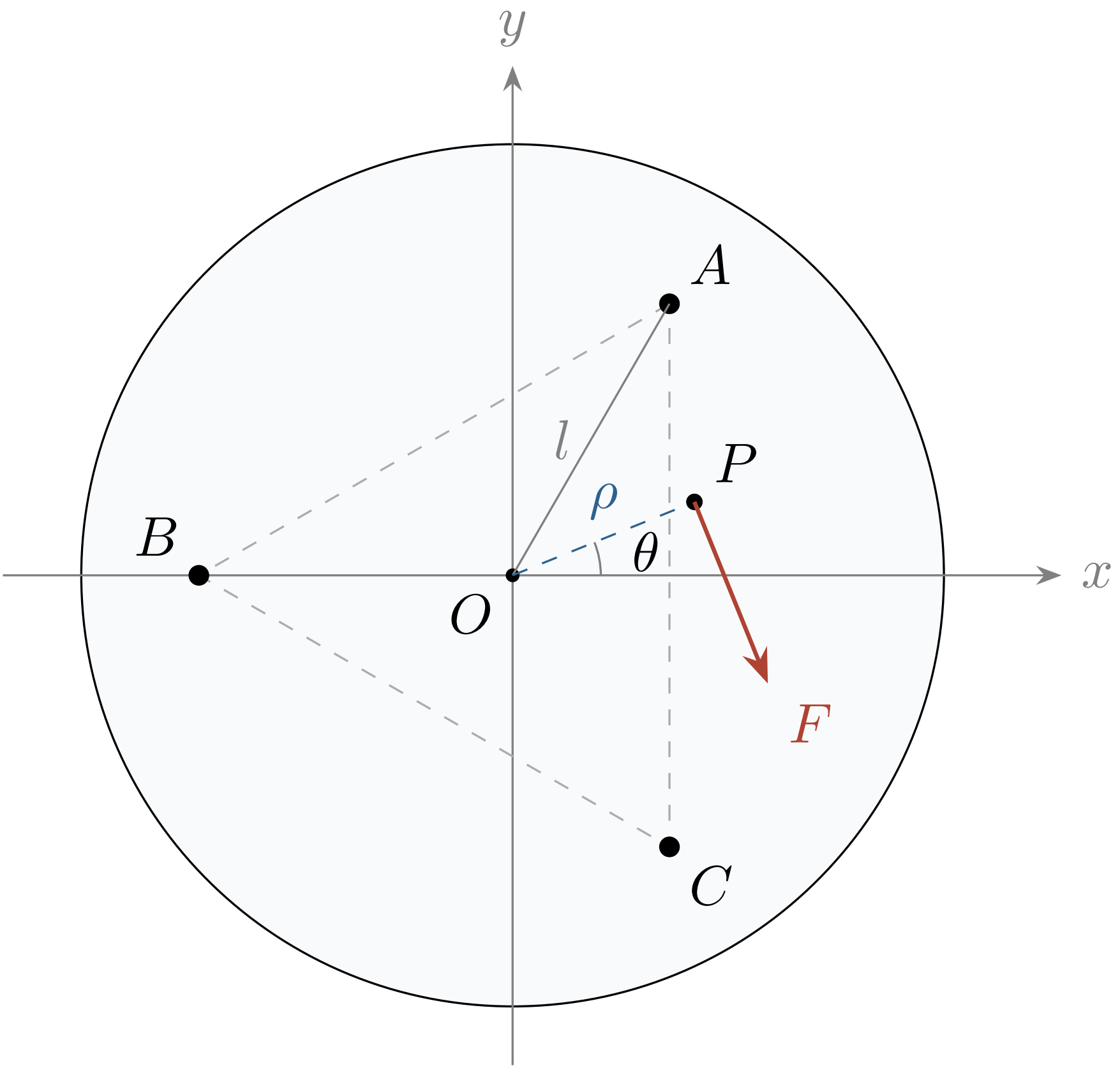
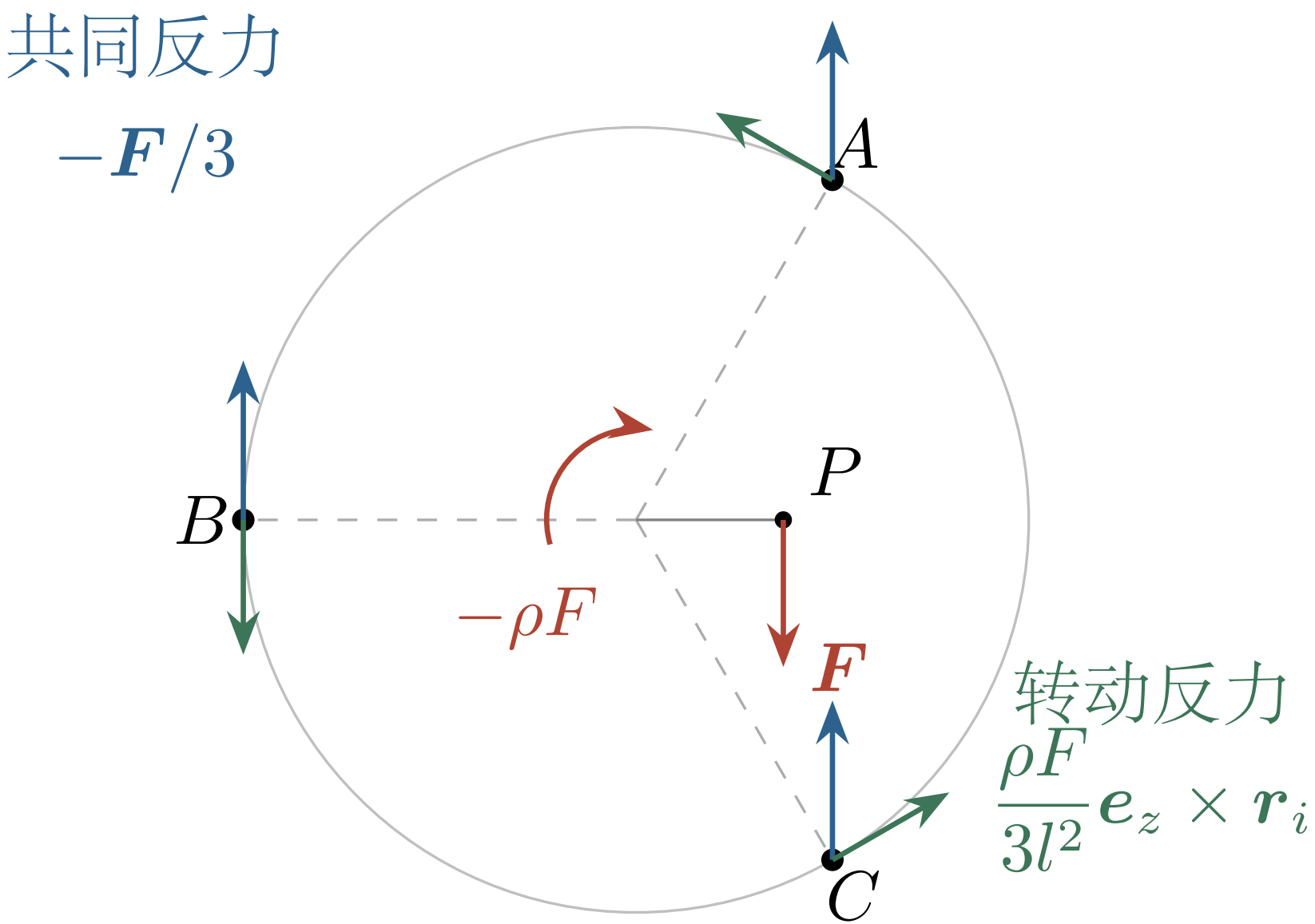
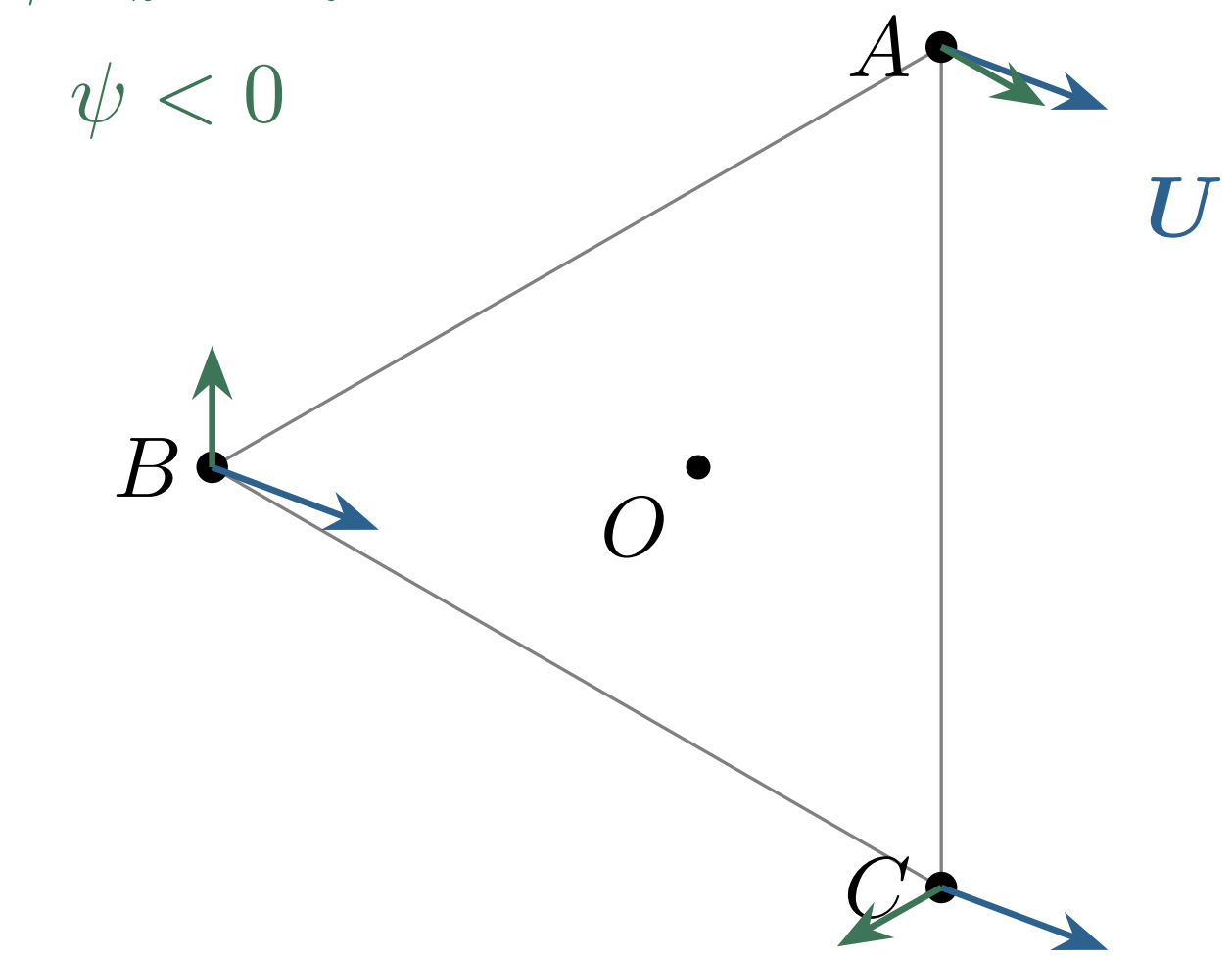


俯视：各腿距 O 为 l；施力沿顺时针切向



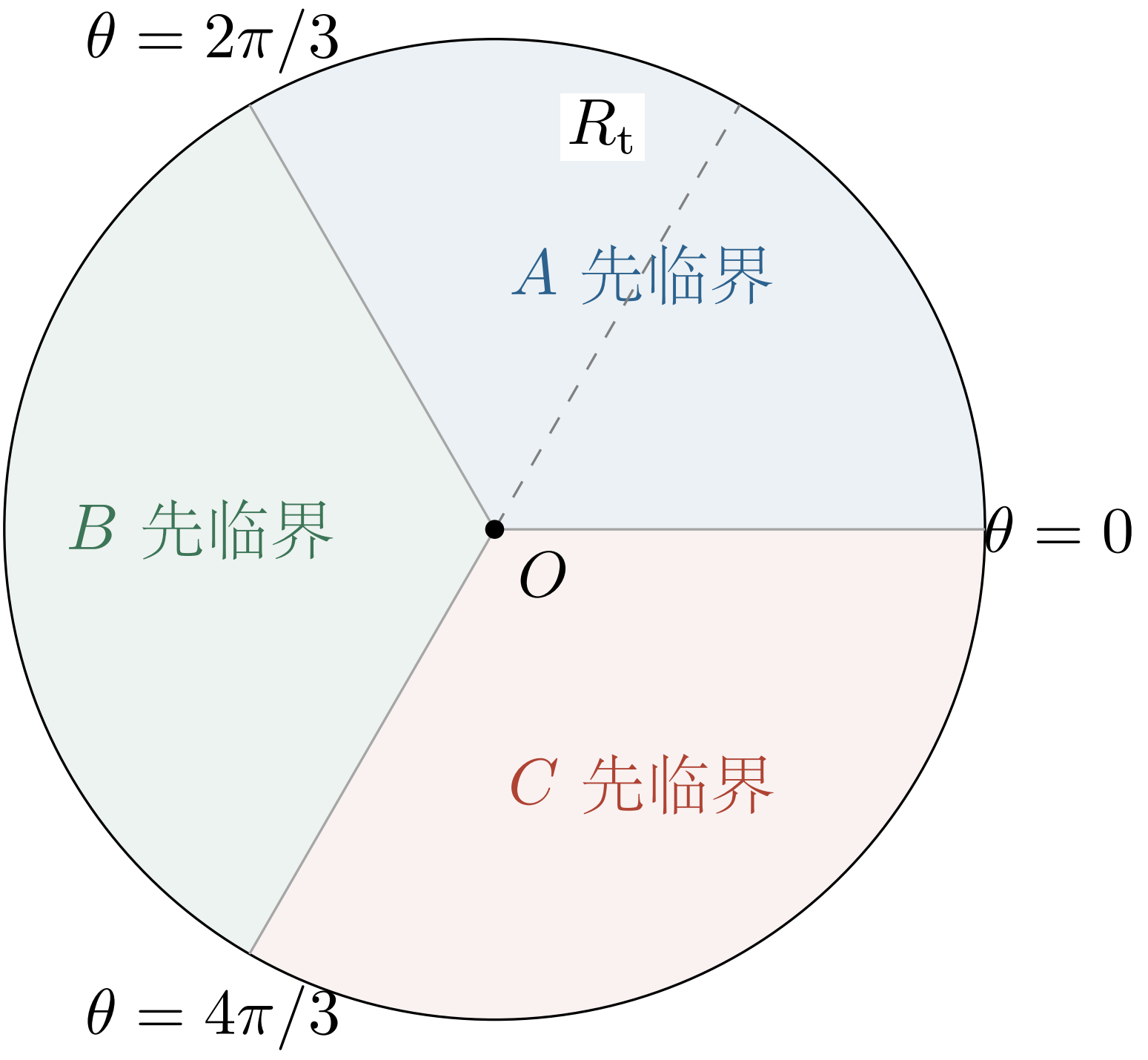
$$\sum_i \boldsymbol{f}_i = -\boldsymbol{F}$$

$$\sum_i (\boldsymbol{r}_i \times \boldsymbol{f}_i)_z = \rho F$$



$$\boldsymbol{u}_i = \boldsymbol{U} + \psi \boldsymbol{e}_z \times \boldsymbol{r}_i$$

$$\boldsymbol{f}_i = -\kappa \boldsymbol{u}_i$$



边界上两腿同时临界；圆心三腿同时临界

条件：三腿相同切向微弹性，固定支持力